Posterior Predictive Check
Model−predicted lines should resemble observed data line
Density
0.02
0.01
0.00
0
100
200
roundElapsed_s
Observed data
Model−predicted data
Linearity
Reference line should be flat and horizontal
Residuals
200
100
0
10
20
30
40
Fitted values
Homogeneity of Variance
Reference line should be flat and horizontal
√|Std. residuals|
4
3
2
1
0
10
20
30
40
Fitted values
Influential Observations
Points should be inside the contour lines
Std. Residuals
2105
1360
95
389
0.8
0.8
200
100
0
0.00
0.02
0.04
0.06
Leverage (hii)
Collinearity
High collinearity (VIF) may inflate parameter uncertainty
Variance Inflation Factor (VIF, log−scale)
50
30
10
5
3
1
coinLabel
coinLabel:t_early_20
coinLabel:t_late_20
coinSentiment
recentSwapRate
t_early_20
t_late_20
Low (< 5)
Moderate (< 10)
High (>= 10)
Normality of Residuals
Dots should fall along the line
Sample Quantile Deviations
200
100
0
−50
−25
0
25
50
Standard Normal Distribution Quantiles
Normality of Random Effects (participantID)
Dots should be plotted along the line
RE Quantiles
10
5
0
−5
−10
−2
−1
0
1
2
Theoretical Quantiles